NANDITHA V MAHESH

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EDUCATION

University of Southern California - Viterbi School of Engineering, Los Angeles, California **Aug. 2022 – May 2024**
M.S., Electrical Engineering 3.8/4.0 GPA
Coursework: EE477 CMOS VLSI Circuit Design, **EE457** Computer System Organization, **EE 577A** VLSI System Design, **EE658** Diagnosis and Design of Reliable Digital Systems

Visvesvaraya Technological University, Belagavi, India **Aug. 2017 – Jul. 2021**
B.E., Electronics & Communication 9.05/10.0 GPA
Relevant Coursework: Digital Electronics, Digital System Design using Verilog, VLSI Design**,** Power Electronics, Nano electronics.

TECHNICAL SKILLS

Programming languages: Verilog HDL, C, Python, JavaScript | **Tools:** Cadence Virtuoso®, ModelSim, Xilinx ISE, Intel Quartus Prime
Hardware: FPGA De0Nano, NVIDIA Jetson Nano

ACADEMIC PROJECTS

512 Bit SRAM with Four 8x16 Banks using Butterfly Architechture: Cadence Virtuoso® **Mar. 2023 – Apr. 2023**

- Achieved a maximum clock frequency of 1.2Ghz & Layout Area of 2679.36 μm², RSNM of 221mV & WSNM 470.73mV at a supply of 1V using gpdk045 on Cadence Virtuoso.
- SRAM consisted of 6T Cell, Precharge Circuitry, Row, Column Decoders, Column Mux, Read/Write Drivers.

5 - stage pipeline design in Verilog HDL: ModelSim **Oct. 2022 – Dec. 2022**

- Designed a 5-stage pipeline early branch & late branch CPU with Hazard Detection & Forwarding Units, with R-type, load word & store word instructions & handling their respective stall dependencies in the pipeline and achieved about 30% reduced CPU time and better performance compared to the pipeline design without Forwarding Unit and Hazard Detection Units.
- Assimilated the concepts of Single Cycle CPU such as data stationary method of control, RAW & WAR dependencies, & their requirements for forwarding or stalling accordingly.

Full System with Round Robin Priority Arbiter for 5-Bit Multiplier & Divider: Cadence Virtuoso® **Oct. 2022 – Dec. 2022**

- Designed & implemented a 5-bit multiplier & divider on Cadence Virtuoso. Furnished a vector file as input & simulated the schematic circuits through Analog Design Environment to verify the circuit.
- Analyzed the FSM for the arbiter circuit to produce a select line to a 2:1 multiplexer to grant or stall the calculated multiplier & divider outputs. Implemented the FSM as a schematic using combinational logic & D flip flops with the reference of FSM & Karnaugh Maps.

16 Bit Ripple Carry Adder (RCA) and Carry Skip Adder (CSA) and layouts: Cadence Virtuoso® **Oct. 2022 – Dec. 2022**

- Constructed the full adder circuit and other standard cells required for the implementation of RCA and CSA. Deployed different optimization techniques to achieve best case delay for sum and carry values in full adder for 13% of area saving.
- Instantiated 16 full adders to arrive at the 16-bit RCA schematic followed by the design of its physical layout. Utilized the same for constructing the 16-bit CSA and its physical layout, which has 9% better performance than 16-bit RCA due to its carry skipping technique.

FPGA De0 Nano testing and programming

- Implemented Verilog code for 4-bit comparator, Binary to ASCII Converter, 8:1 Mux and sequence detector successfully programmed FPGA De0 Nano board for their deployment, using the Intel Quartus Prime tool.

WORK EXPERIENCE

Software Engineer, Alltramatic, New York, USA (remote) **Aug. 2021 – Jul. 2022**

- Designed front-end application using ReactJS & Google Maps APIs to help the property owners mark & edit different access points on their lawn for the navigation of an Automated Ground Vehicle (AGV) whose main processor was NVIDIA Jetson Nano board.
- Marked outer boundaries for over 100 zip-codes & 10,000 properties on Google Maps embed on the website. Stored those coordinates in a Firebase database. Streamlined the different user specific geo-coordinates of the lawn's different landmarks and prime locations marked on the webpage to bolster easy automation & overall navigation of the AGV using web development.

Intern, Alltramatic, New York, USA (remote) **Jan. 2021 – Jun. 2021**

- Self-taught different OpenCV modules & Python programming skills to improve pictures of lawn properties for implementation of AGV object detection feature. Managed the path planning, navigation design, & testing for a lawnmower Automated Ground Vehicle (AGV).
- Formulated the input points from Firebase database to AGV for testing under different conditions thereby increased the drive performance of AGV by **25%** for wide range of inputs.

CERTIFICATIONS

- *Basics of Static Timing Analysis v2.0***:** Cadence Design Systems **Apr. 2023**

PUBLICATIONS

- V. M., Nanditha, et al., *"Design and Analysis of Digital Circuits using Quantum Cellular Automata and Verilog."* 2020 7th International Conference on Computing for Sustainable Global Development (INDIACom), May 2020.